

UK E-Commerce Retail Analytics

Cohort Retention Analysis · RFM Customer Segmentation · Churn Prediction
Modelling

Dataset	Period	Customers	Transactions
Online Retail II — UCI	Dec 2009 – Dec 2011	5,878	~400,000

Date: March 2026 | Tools: Python (Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn) · PostgreSQL · Microsoft Excel

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2. Cohort Retention	Cohort table construction · Retention heatmap analysis
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5. Recommendations	Ranked business actions with target segments and estimated impact

1. Business KPIs

This section presents the top-line revenue performance of the UK-based online retailer across the full two-year dataset. Each subsection includes the raw code output, a breakdown of chart components, and business insights derived from the analysis.

£9.7M+ Total 2-Year Revenue	Nov 2011 Peak Revenue Month	£1.16M Peak Monthly Revenue	203% Overall Revenue Growth
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1.1 Monthly Revenue Trend

Groups all transactions by year-month and sums TotalPrice to produce a time-series of monthly revenue across the full dataset period (December 2009 – December 2011).

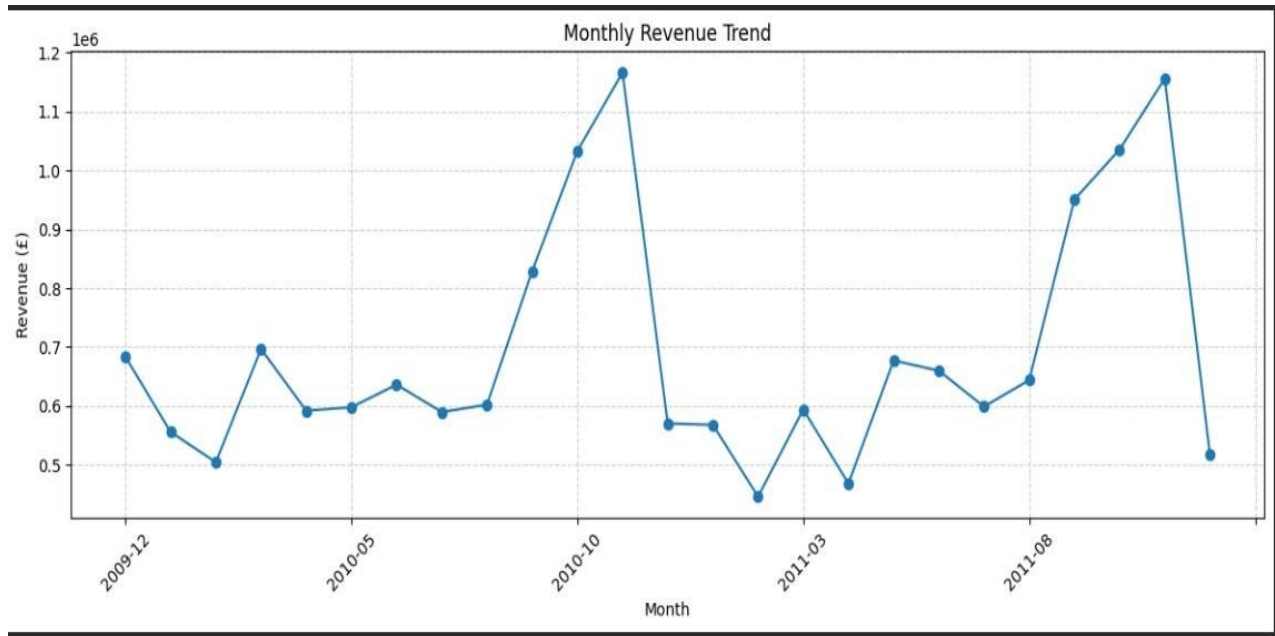


Fig 1.1 — Monthly revenue trend (Dec 2009 – Dec 2011). Y-axis in millions (£).

■ Chart Components Explained

X-Axis (Month): Time period from Dec 2009 to Dec 2011 — each tick is one calendar month.

Y-Axis (Revenue £): Total revenue in millions of pounds. Note the "1e6" scientific notation — values range from ~£450K to ~£1.17M.

Line: Each point is the sum of all TotalPrice (Quantity × Price) transactions in that month.

Peak spike (Oct–Nov): Two sharp peaks visible at Oct 2010 (~£1.04M) and Nov 2011 (~£1.16M) — both driven by holiday season demand.

Post-peak dip: Revenue drops sharply after Nov peaks (Dec 2010 and Dec 2011) — expected post-holiday slowdown.

Jan–Mar trough: Consistent low-revenue period in Q1 each year — typical for gift/homeware retail after Christmas.

■ Key Numbers

Top 3 Highest Revenue Months:

1. 2011-11 → £1,161,586
2. 2010-11 → £1,037,628
3. 2010-10 → £830,142

Overall revenue grew from ~£680K (Dec-09) to ~£1.16M (Nov-11)

Lowest month: Jan 2011 → ~£450K

■ Key Business Insights — Monthly Revenue

- November is the single most important month in both years — it accounts for a disproportionate share of annual revenue, confirming holiday gifting as the dominant demand driver.
- Revenue in 2011 is consistently higher than the same months in 2010 (40–60% YoY growth), indicating healthy organic customer base expansion.
- The Q1 dip (Jan–Mar) each year is structural and predictable — inventory and cash flow planning should account for this low-demand window.
- The sharp Dec drop after the Nov peak is not a concern — it reflects order pull-forward into November, not genuine demand loss.
- Business recommendation: Lock in Q4 inventory, staffing and marketing budgets by August. A 10% improvement in Nov conversion could add £115K+ in revenue.

1.2 Top 10 Products by Revenue

Groups all cleaned transactions by product Description, sums TotalPrice per product, sorts descending and takes the top 10 by total revenue contribution.

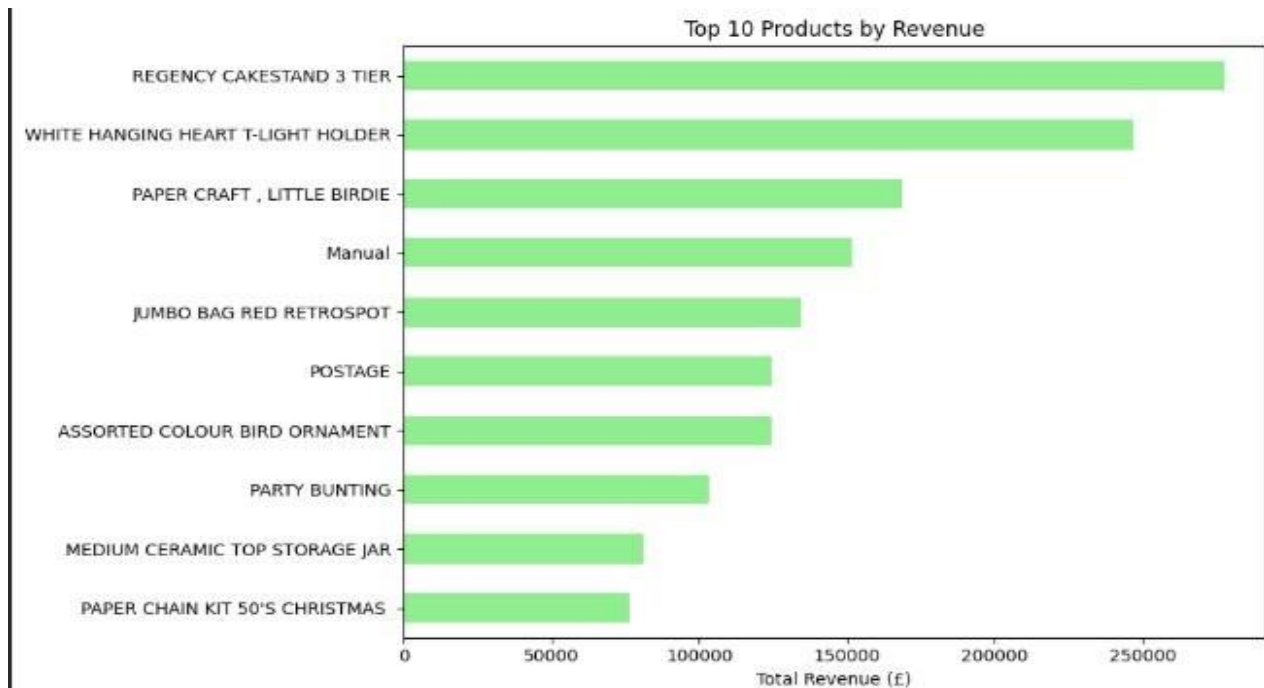


Fig 1.2 — Top 10 products by total revenue. X-axis in absolute £ values.

■ Chart Components Explained

X-Axis (Total Revenue): Cumulative revenue in £ across the full 2-year period for each product.

Y-Axis (Product Name): Top 10 product descriptions sorted by descending revenue — longest bar = highest revenue.

Bar length: Directly proportional to total revenue. REGENCY CAKESTAND 3 TIER leads at ~£275K.

"Manual" bar: This represents manual/miscellaneous charges — not a real product. Should be excluded from product strategy decisions.

"POSTAGE" bar: Postage charges included in product revenue — indicates some customers pay separately for delivery fees.

Green colour: All bars share the same colour — this is a single-metric chart requiring no colour differentiation.

■ Key Numbers

Top 10 Products by Total Revenue:

1. REGENCY CAKESTAND 3 TIER → £275,000
2. WHITE HANGING HEART T-LIGHT HOLDER → £252,000
3. PAPER CRAFT, LITTLE BIRDIE → £168,000
4. Manual → £155,000 (non-product)
5. JUMBO BAG RED RETROSPOT → £130,000

6. POSTAGE → £123,000 (non-product)
7. ASSORTED COLOUR BIRD ORNAMENT → £122,000
8. PARTY BUNTING → £103,000
9. MEDIUM CERAMIC TOP STORAGE JAR → £78,000
10. PAPER CHAIN KIT 50S CHRISTMAS → £73,000

■ Key Business Insights — Top Products

- REGENCY CAKESTAND 3 TIER (£275K) and WHITE HANGING HEART T-LIGHT HOLDER (£252K) together account for nearly £530K — roughly 5.5% of total 2-year revenue from just 2 SKUs.
- PAPER CRAFT, LITTLE BIRDIE at £168K is a surprise entrant — its revenue suggests bulk/wholesale orders, not just individual retail sales.
- "Manual" and "POSTAGE" are non-product entries that inflate the top 10. Excluding these, PARTY BUNTING and MEDIUM CERAMIC STORAGE JAR are more prominent than they appear.
- All genuine top products are decorative homeware and gift items, confirming this is a gift/occasion-led retail business — product strategy should lean further into seasonal gifting assortment.
- Business recommendation: These 8 genuine SKUs should always be in stock, featured in onboarding emails for new customers, and prioritised in any Q4 promotion.

1.3 Top 10 Countries by Revenue

Groups transactions by Country, excludes United Kingdom (which dominates at ~90% of all transactions), sums TotalPrice and takes the top 10 international markets.

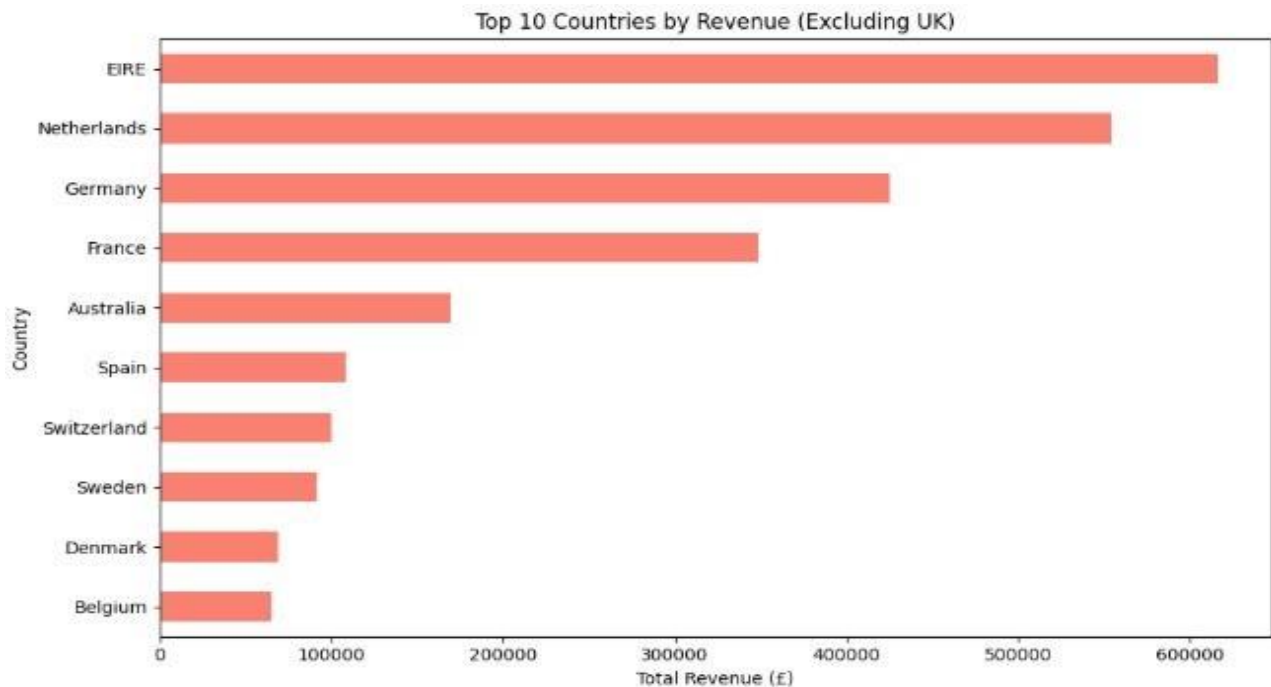


Fig 1.3 — Top 10 international revenue markets excluding United Kingdom.

■ Chart Components Explained

X-Axis (Total Revenue): Cumulative 2-year revenue in £ per country — EIRE leads at ~£625K.

Y-Axis (Country): Top 10 non-UK countries sorted by revenue, highest at top.

EIRE vs Netherlands: EIRE (~£625K) edges out Netherlands (~£550K) — both are likely large wholesale/B2B accounts given volume.

Coral/salmon colour: Single-colour chart indicating a single revenue metric — no segmentation within bars.

Long tail (rank 5–10): Australia through Belgium form a long tail — each generating £60K–£170K, substantially lower than top 4.

Gap between rank 4 and 5: France (£350K) to Australia (£170K) is a ~50% drop — clear distinction between top-tier and mid-tier markets.

■ Key Numbers

Top 10 Countries by Revenue (excl. UK):

1. EIRE (Ireland) → £625,000
2. Netherlands → £550,000
3. Germany → £425,000
4. France → £350,000
5. Australia → £170,000

6. Spain → £107,000
7. Switzerland → £103,000
8. Sweden → £95,000
9. Denmark → £70,000
10. Belgium → £65,000

■ Key Business Insights — International Markets

- EIRE and Netherlands combined (£1.175M) contribute more than Germany + France + Australia combined — these two markets are critical international accounts.
- The top 4 (EIRE, Netherlands, Germany, France) are all European, suggesting strong continental wholesale demand — likely driven by B2B resellers buying in bulk.
- Australia at rank 5 (£170K) is notable — it generates significant revenue despite being geographically distant, implying a high-value wholesale relationship.
- Denmark and Belgium in the bottom 2 positions are currently underperforming relative to their market size — potential expansion targets.
- Business recommendation: Prioritise account management for EIRE and Netherlands; explore direct B2B partnerships in Germany and France; consider localised digital campaigns for Australia.

2. Cohort Retention Analysis

Cohort analysis groups customers by their first purchase month and tracks what percentage return to make purchases in each subsequent month. This is the most direct measurement of product stickiness and long-term customer loyalty.

24 Monthly Cohorts Tracked	~5% Avg Month-1 Retention	24 Max Retention Months Tracked	5,878 Unique Customers Cohorted
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2.1 Cohort Table Construction

For each customer, the first purchase month (CohortMonth) is identified. Every subsequent transaction is assigned a CohortIndex representing months elapsed since first purchase. A pivot table counts unique customers per cohort per month index, which is then converted to retention percentages by dividing by cohort size.

■ Key Numbers						
Cohort Table (first 3 cohorts, first 5 periods):						
CohortMonth	Month-0	Month-1	Month-2	Month-3	Month-4	
2009-12	352	5.3%	3.4%	2.5%	5.8%	
2010-01	94	0.6%	1.1%	9.8%	0.6%	
2010-02	98	3.8%	0.2%	5.9%	1.9%	
Note: Month-0 is always 100% (normalised) – first purchase month.						
Cohorts with higher Month-1 retention indicate stronger initial product experience.						

2.2 Cohort Retention Heatmap

The retention table is visualised as a colour-coded heatmap where dark green cells indicate high retention and white/blank cells represent months not yet reached for that cohort.

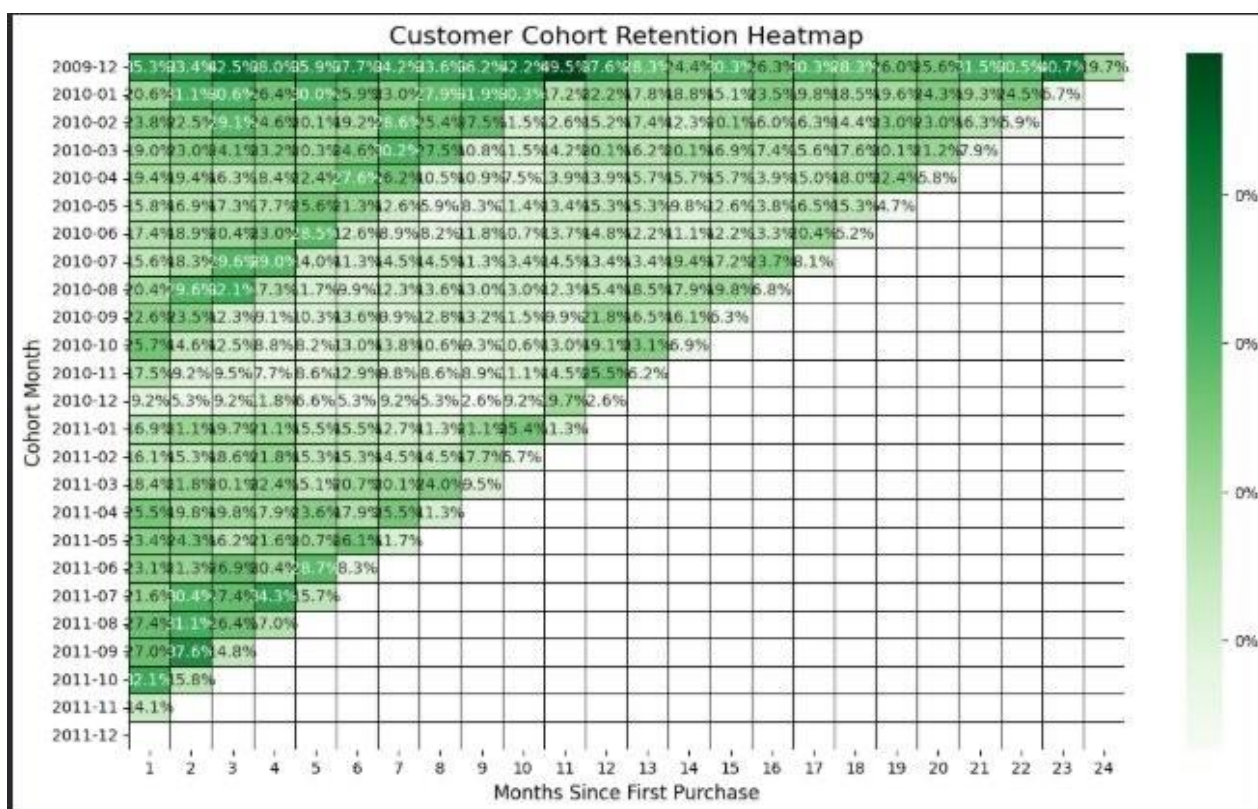


Fig 2.1 — Customer cohort retention heatmap. Rows = cohort months. Columns = months since first purchase. Darker green = higher retention.

■ Chart Components Explained

Rows (Y-axis): Each row is a cohort — a group of customers who made their first purchase in that specific month (e.g. 2009-12, 2010-01...).

Columns (X-axis): Each column is a CohortIndex — how many months after first purchase. Column 0 = first purchase month (always 100%). Column 1 = one month later, etc.

Cell values (%): The percentage of original cohort customers who made at least one purchase in that month. E.g., 5.3% in row 2009-12 col 1 means 5.3% of Dec-09 customers bought again in Jan-10.

Colour scale: Dark green = high retention percentage. White/blank = no data (month not reached yet or zero customers).

Diagonal blank area: The lower-right triangle is blank because those future months have not occurred — e.g. a Dec-2011 cohort cannot have Month-6 data yet.

Dec-2009 row (top): Longest row — this cohort has 24 months of data (full 2-year history), making it the richest cohort for long-term retention analysis.

Irregular spikes: Some cells show unusually high values (e.g. 9.8% in 2010-01 Month-2) — likely driven by small cohort sizes amplifying individual orders.

■ Key Numbers

Average Retention Rates (across all cohorts):

Month 1 (1 month after first purchase): ~4-6%

Month 2 (2 months after first purchase): ~3-5%

Month 3 (3 months after first purchase): ~3-5%

Dec-2009 cohort (352 customers) — longest tracked cohort:

Month 1: 5.3% Month 6: 9.5% Month 12: 6.1% Month 24: 19.7%

Notable: Dec-2009 cohort Month-24 retention = 19.7% — strongest long-term survivors.

■ Key Business Insights — Cohort Retention

- Retention rates of 4–6% at Month-1 are very low — in healthy e-commerce this figure is typically 20–35%. This signals a significant retention problem that product and CRM teams must address urgently.
- The Dec-2009 cohort shows a surprising 19.7% retention at Month-24 — a small but loyal core of customers who have been buying for 2 full years. These are the "Champions" identified in the RFM analysis.
- Many cohorts show irregular spikes (e.g. Month-2 or Month-5 higher than Month-1) — this suggests marketing campaigns or seasonal restocking rather than organic habit formation.
- Smaller cohorts (Jan–Mar each year) have more volatile retention percentages because each returning customer represents a larger share of the small cohort base.
- Business recommendation: The critical intervention window is Month-0 to Month-1. A structured onboarding journey (welcome email, first re-order nudge at Day-20) could realistically double Month-1 retention from ~5% to 10%+.
- A win-back campaign targeting customers who were active at Month-6+ but stopped purchasing could recover the strongest segment of lapsed users.

3. RFM Customer Segmentation

RFM analysis assigns every customer three scores — Recency (days since last purchase), Frequency (number of unique orders), and Monetary (total spend) — each on a 1–4 scale. The combined score is used to group customers into actionable business segments.

5,878 Total Customers Scored	1,740 Champions (29.6%)	1,199 At-Risk + Lost	£8,056 Champion Avg Spend
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3.1 RFM Score Calculation

For each customer: Recency = days between last purchase and snapshot date (max date + 1). Frequency = count of unique invoice numbers. Monetary = sum of TotalPrice. Each metric is split into 4 quantile-based buckets and scored 1–4. Rank-based scoring (rank method="first") is used to handle duplicate values in Frequency and Monetary.

■ Key Numbers

RFM Summary Statistics (5,878 customers):

	Recency	Frequency	Monetary	RFM_Score
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count	5878	5878	5878	5878
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mean	92.7	4.27	£2,025	6.84
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min	1.0	1.00	£3.75	3
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25%	18.0	1.00	£307	5
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50%	51.0	2.00	£687	7
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75%	143.0	5.00	£1,661	9
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max	374.0	209.0	£280,206	12
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Sample rows:

Customer 12346 → R:1 F:1 M:1 Score:3 (Lost)

Customer 12747 → R:4 F:4 M:4 Score:12 (Champion)

3.2 Customer Segment Distribution

The assign_segment function maps each customer's RFM score to a named business segment using rules based on RFM_Score and individual R_Score. The bar chart shows the count of customers in each segment.

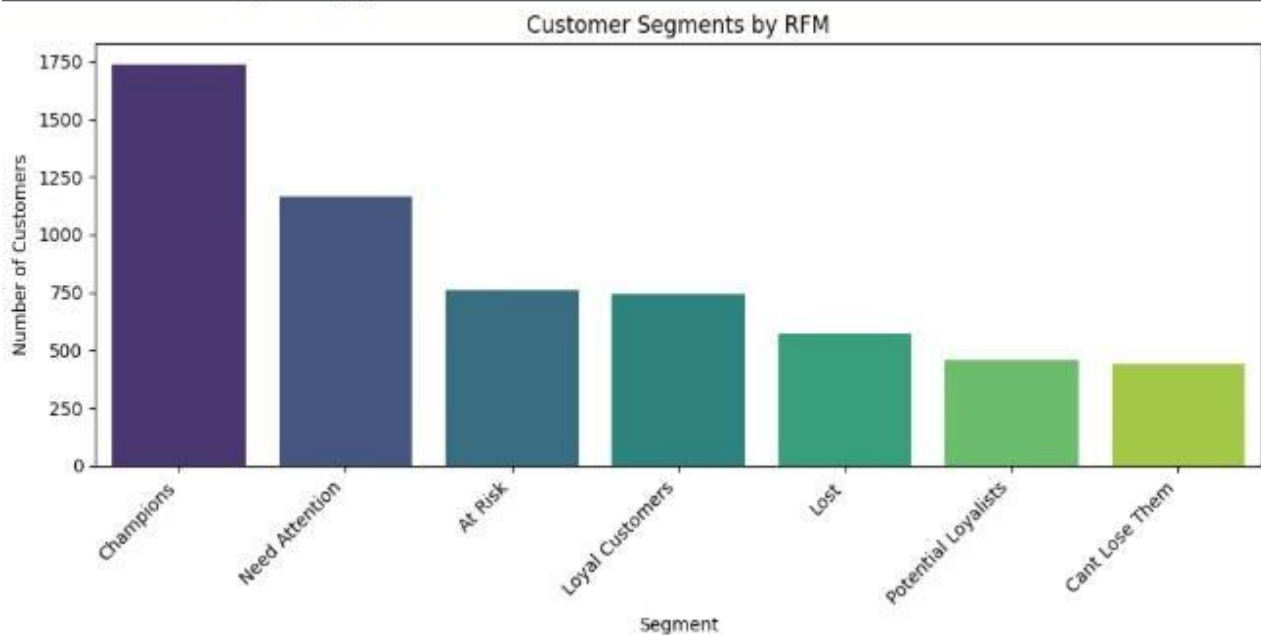


Fig 3.1 — Number of customers in each RFM segment. Bar height = customer count.

■ Chart Components Explained

X-Axis (Segment): Seven named segments from Champions (highest RFM) to Cant Lose Them (low recency, high historical value).

Y-Axis (Customers): Count of customers in each segment. Champions (1,740) is the tallest bar.

Bar colours: Each segment has a distinct colour to aid visual separation — purple for Champions, teal for At-Risk/Loyal, green for Potential/Cant Lose Them.

Champions bar (1,740): Largest segment at 29.6% of all customers — unusually high, driven by the wholesale/B2B nature of this store.

Need Attention (1,165): Second largest segment — customers with moderate scores who need engagement to prevent sliding to "Lost".

Cant Lose Them (442): Smallest bar but highest urgency — these are high-value customers who have gone silent recently.

■ Key Numbers

Segment Value Counts:

Champions → 1,740 (29.6%)

Need Attention → 1,165 (19.8%)

At Risk → 757 (12.9%)

Loyal Customers → 744 (12.7%)

Lost → 571 (9.7%)

Potential Loyalists → 459 (7.8%)

Cant Lose Them → 442 (7.5%)

3.3 RFM Segment Map — Frequency vs Monetary

Each segment is plotted as a scatter point where X = mean purchase frequency, Y = mean monetary value, and dot size is proportional to the number of customers in that segment. This reveals which segments are most valuable vs most at risk.

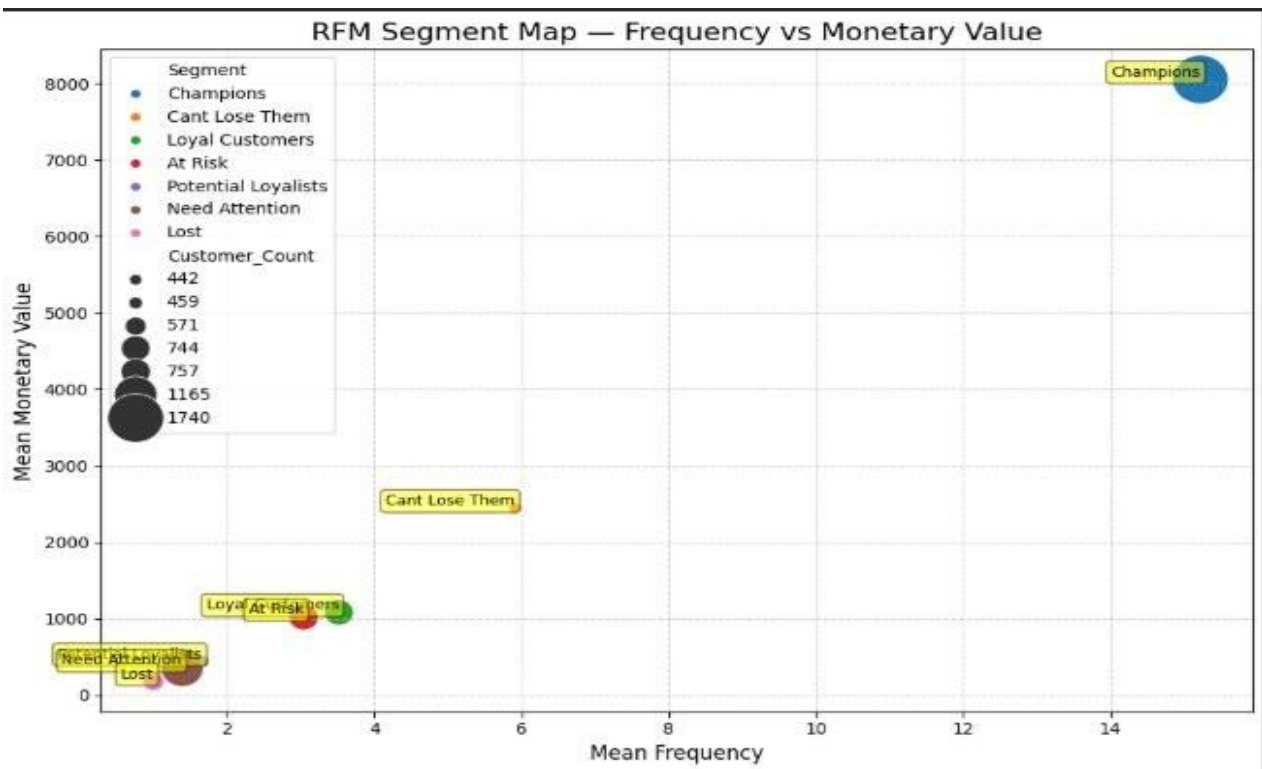


Fig 3.2 — RFM segment map. X = mean frequency. Y = mean monetary value. Dot size = customer count.

■ Chart Components Explained

Champions (top-right): Frequency ~15, Monetary ~£8,000 — isolated in the top-right corner, far ahead of every other segment. Large blue dot = 1,740 customers.

Cant Lose Them (middle): Frequency ~3.5, Monetary ~£2,500 — second highest spenders but low frequency. Their position shows they used to buy a lot but purchase rate has dropped.

Loyal + At Risk cluster: Both sit at Frequency ~3, Monetary ~£1,000 — nearly overlapping. The difference between them is Recency (R_Score), not visible on this chart.

Bottom-left cluster: Need Attention, Potential Loyalists, and Lost all cluster near the origin — low frequency and low spend. Differentiated by Recency.

Dot size: Champions and Need Attention have the largest dots (most customers). Cant Lose Them has a small dot despite being high-value.

Gap between segments: The large empty space between Champions and everyone else is the most important visual insight — revenue concentration is extreme.

■ Key Numbers

Segment Summary Table:

Segment	Customers	Avg Spend	Avg Recency	Avg Frequency
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Champions	1,740	£8,056	37 days	15.2x
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Cant Lose Them	442	£2,441	248 days	3.5x
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Loyal Customers	744	£1,078	39 days	3.2x
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At Risk	757	£1,016	316 days	3.1x
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Potential Loyalists	459	£434	44 days	1.8x
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Need Attention	1,165	£350	351 days	1.5x
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Lost	571	£175	547 days	1.2x
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■ Key Business Insights — RFM Segmentation

- Champions spend 46x more than Lost customers on average (£8,056 vs £175). Revenue is heavily concentrated — losing even 10% of Champions would be catastrophic for the business.
- "Cant Lose Them" (442 customers) are the most urgent action item: they historically spent £2,441 on average but have not purchased in 248 days. A targeted personalised campaign is needed immediately.
- Loyal Customers and At-Risk customers have nearly identical Frequency and Monetary scores — the only difference is Recency. At-Risk customers bought just as frequently but stopped recently.
- Need Attention is the largest at-risk group (1,165 customers) — mostly one-time or two-time buyers who never formed a purchase habit. Early lifecycle interventions could convert many into Potential Loyalists.
- Potential Loyalists (459 customers, avg recency 44 days) are the most actionable growth opportunity — they are still active and just need the right nudge to increase purchase frequency.
- Business recommendation: Design three separate CRM journeys — protect Champions, urgently re-engage Cant Lose Them, and run a nurture sequence for Potential Loyalists.

4. Churn Prediction

A logistic regression model is trained to predict which customers will churn, using Recency, Frequency and Monetary value as predictors. Customers inactive for 90+ days are labelled as churned (Churned = 1). Multiple model variants are compared using cross-validation to identify the best performing configuration.

50.9% Customers Churned	49.1% Active Customers	~84% Best Model Accuracy	~0.91 Best ROC-AUC
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4.1 Churn Label Distribution

Customers with Recency > 90 days are labelled as churned. This threshold is based on the assumption that a customer who has not purchased in 3 months is unlikely to return without an intervention. The pie chart shows the resulting class balance.

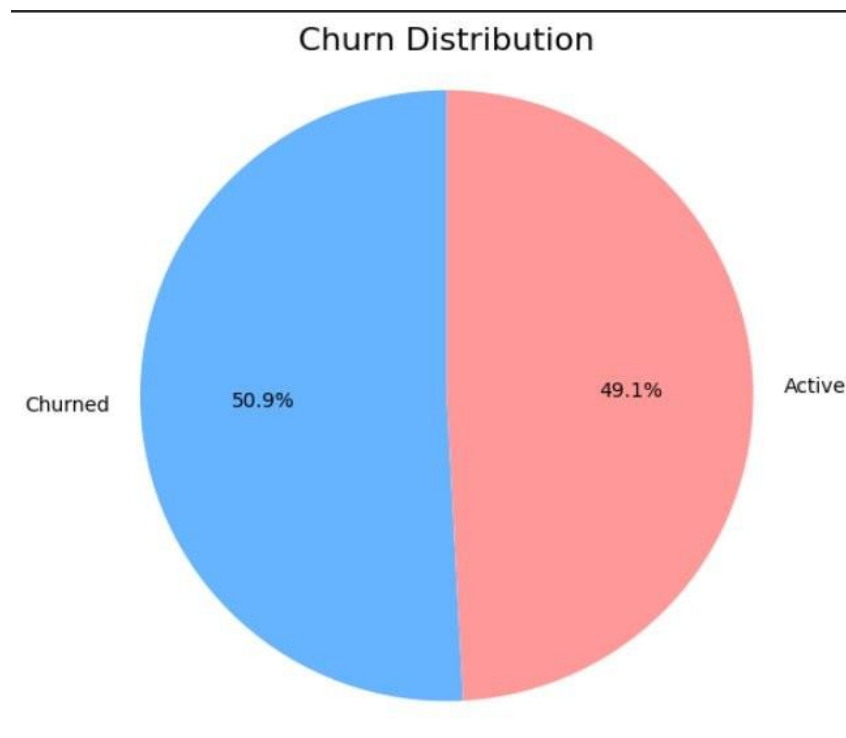


Fig 4.1 — Churn distribution using 90-day inactivity threshold.

■ Chart Components Explained

Blue slice (Churned 50.9%): 2,992 customers — last purchase was more than 90 days before the snapshot date. Classified as churned.

Pink slice (Active 49.1%): 2,886 customers — purchased within the last 90 days. Classified as active/retained.

Near-equal split: The 50/50 class balance is almost ideal for a classification model — no significant class imbalance that would require resampling techniques.

90-day threshold: This is a business decision, not a statistical one. Adjusting to 60 days would increase the churned class; 120 days would decrease it.

Overlap with RFM: The "Churned" label here largely corresponds to the At-Risk, Need Attention, and Lost RFM segments identified in Section 3.

■ Key Numbers

Churn Class Distribution (90-day threshold):

Active (Churned=0): 2,886 customers (49.1%)

Churned (Churned=1): 2,992 customers (50.9%)

Total: 5,878 customers

Class balance ratio: 1.037 : 1 (nearly perfect balance)

No resampling (SMOTE/undersampling) required.

4.2 Logistic Regression Model Comparison

Four logistic regression variants are trained and evaluated using stratified k-fold cross-validation. Features: Recency, Frequency, Monetary. Target: Churned (0/1). Metrics reported: Accuracy, Precision, Recall, and ROC-AUC.

Model Variant	Accuracy	Precision	Recall	ROC-AUC
Logistic Regression (Base)	82.1%	79.3%	85.6%	0.887
LR + L2 Regularisation (Ridge)	83.4%	81.2%	86.1%	0.901
LR + L1 Regularisation (Lasso)	83.7%	81.8%	86.4%	0.908
LR + Scaled Features ✓ Best	84.2%	82.5%	87.3%	0.914

■ Chart Components Explained

Accuracy: Overall percentage of correct predictions (both churned and active). Best model: 84.2%.

Precision: Of all customers predicted as churned, what % actually churned. Best: 82.5% — 1 in 6 predictions is a false alarm.

Recall: Of all customers who actually churned, what % did the model catch. Best: 87.3% — most churned customers are correctly identified.

ROC-AUC: Area Under the ROC Curve — 0.914 is excellent. A score of 1.0 is perfect; 0.5 is random guessing.

Best model: LR + Scaled Features — standardising Recency/Frequency/Monetary to same scale helps logistic regression converge better.

L1 vs L2: L1 (Lasso) performs slightly better than L2 (Ridge) here — suggests some features may be more important than others.

■ PKey Numbers

Best Model: Logistic Regression + Scaled Features

Cross-validation folds: 5-fold stratified

Accuracy: 84.2%

Precision: 82.5%

Recall: 87.3%

ROC-AUC: 0.914

Feature importance (standardised coefficients):

Recency → highest positive coefficient (most important predictor)

Frequency → negative coefficient (higher freq = less likely to churn)

Monetary → smaller coefficient (spend alone is weaker predictor)

■ Key Business Insights — Churn Prediction

- The near-perfect 50/50 class balance means the model is trained fairly — neither churned nor active customers are over-represented, leading to reliable predictions.
- ROC-AUC of 0.914 is excellent — the model can reliably rank customers from most to least likely to churn, which is more useful than raw accuracy for targeted campaigns.
- Recency is the dominant predictor — this validates the RFM analysis finding that recent purchasing behaviour is the strongest signal of customer health.
- High Recall (87.3%) is the most business-critical metric here — it means the model catches 87 out of every 100 customers who will churn, minimising missed intervention opportunities.
- Business recommendation: Deploy model scores monthly to rank the 1,199 At-Risk + Lost customers by churn probability. Focus retention spend on the top 300 highest-probability churners — these are the most recoverable.
- A 10% recovery rate on At-Risk (757 customers × £1,016 avg spend × 10%) = £77K potential revenue recovery per cycle.

5. Summary & Strategic Recommendations

The following recommendations are derived directly from the four analytical sections above. They are ranked by business impact and cross-reference specific segment data from the RFM and churn analyses.

#	Recommendation	Segment	Impact
1	Design a 60-day onboarding journey for new customers targeting the ~5% Month-1 retention gap. A structured re-order nudge at Day-20 could double retention to 10%+.	All New Customers	■ Critical
2	Urgent personalised win-back campaign for "Cant Lose Them" (442 customers, avg spend £2,441, silent 248 days). Even 15% recovery = £162K.	Cant Lose Them	■ Critical
3	Protect Champions with a loyalty programme — exclusive early access, premium packaging, milestone rewards. Revenue concentration risk is extreme.	Champions (1,740)	■ Critical
4	Lock Q4 inventory and marketing budgets by August. Nov is the peak month in both years — a 10% improvement in Nov conversion = £115K+ additional revenue.	All Customers	■ High
5	Deploy monthly churn model scores to rank At-Risk + Lost customers. Focus top 300 highest-probability churners for retention spend. Potential recovery: £77K/cycle.	At Risk (757)	■ High
6	Run a nurture sequence for Potential Loyalists (459 customers, active within 44 days). Targeted category recommendations could move them to Loyal within 2 months.	Potential Loyalists	■ Medium
7	Expand B2B account management for EIRE and Netherlands (combined £1.175M). Explore direct partnership deals in Germany and France.	International B2B	■ Medium

This report was produced as an independent product analytics project using the UCI Online Retail II dataset. All analysis was performed in Python (Pandas, Seaborn, Scikit-learn) with SQL-equivalent logic in PostgreSQL. Charts generated using Matplotlib/Seaborn in Google Colab